

Methods

Cell culture

The TR14 neuroblastoma cell line was a gift from J. J. Molenaar (Princess Máxima Center for Pediatric Oncology). Cell line identity for the master stock was verified by STR genotyping (IDEXX BioResearch). The GBM39-KT cell line was derived from a patient with glioblastoma undergoing surgery at Mayo Clinic, Rochester, Minnesota as described previously⁵⁵. Monoclonal spheroids were isolated from GBM39-KT cells by limiting dilution to generate GBM39-KT-D10. The CA718 cell line was derived from a patient with glioblastoma as described previously⁵ and was obtained from the University of California San Diego Moores Cancer Center. Parental SNU16, COLO 320DM, H716 and HCT116 cells were obtained from ATCC. The monoclonal SNU16m1 was a sub-line of the parental SNU16 cells generated from a single cell after lentiviral transduction and stable expression of dCas9-KRAB as we previously described⁴. SNU16 and SNU16m1 cells were maintained in Dulbecco's modified Eagle's medium/nutrient mixture F-12 (DMEM/F12 1:1; Gibco, 11320-082), 10% fetal bovine serum (FBS; Hyclone, SH30396.03) and 1% penicillin–streptomycin (Thermo Fisher Scientific, 15140-122). COLO 320DM cells were maintained in DMEM (Thermo Fisher Scientific, 11995073) supplemented with 10% FBS and 1% penicillin–streptomycin. GBM39-KT cells were maintained in DMEM/F12 1:1, B-27 supplement (Gibco, 17504044), 1% penicillin–streptomycin, GlutaMAX (Gibco, 35050061), human epidermal growth factor (EGF, 20 ng ml⁻¹; Sigma-Aldrich, E9644), human fibroblast growth factor (FGF, 20 ng ml⁻¹; Peprotech) and heparin (5 µg ml⁻¹; Sigma-Aldrich, H3149-500KU). TR14 cells were grown in RPMI 1640 with 20% FBS and 1% penicillin–streptomycin. For the mitotic cell imaging experiments in Fig. 2, SNU16m1 cells were grown in RPMI 1640 with 10% FBS. H716 cells were grown in ATCC formulated RPMI 1640 (Gibco, A1049101) with 10% FBS and 1% penicillin–streptomycin–glutamine. COLO 320DM cells used for live-cell imaging, PC3 and HCT116 were cultured in DMEM (Corning, 10-013-CV) with 10% FBS and 1% penicillin–streptomycin–glutamine. All cells were cultured at 37 °C with 5% CO₂. All cell lines tested negative for mycoplasma contamination.

Chemicals

BRD4 bivalent degrader was a gift from M. M. Hassan and N. S. Gray, and was resuspended in DMSO as 10 mM stock⁵⁶. Triptolide (Millipore, 645900) was resuspended with DMSO as 55 mM stocks and were used at a final concentration of 10 µM. Actinomycin D (Millipore Sigma, SBR00013) was used at a final concentration of 5 µg ml⁻¹. DRB (Sigma-Aldrich, D1916) was resuspended with DMSO as 70 mM stocks and was used at a final concentration of 200 µg ml⁻¹. ZL-12A was synthesized as reported previously⁴⁶ and resuspended in DMSO as 20 mM stock, and was used at a final concentration of 50 µM for 3 h. In the pretreatment assay with triptolide, ZL-12A was added for 3 h, followed by a wash-off with 1× PBS and the addition of DMSO or triptolide (10 µM) for 3.5 h.

Genetic knockout of *CIP2A*

CIP2A-knockout cells were created using the SNU16m1 cells as follows. We designed a guide RNA sequence targeting the protein-coding region of *CIP2A* using CHOPCHOP⁵⁷ (<https://chopchop.cbu.uib.no>), as well as a non-targeting control sgRNA (guide sequences are provided in Supplementary Table 1). To deliver each guide with CRISPR–Cas9 into cells, we mixed purified *S. pyogenes* Cas9 nuclease (Alt-RS.p. Cas9 Nuclease V3; IDT, 1081058) with each single-guide RNA (sgRNA; diluted to 30 µM in 1× TE buffer; Synthego) at a 1:6 molar ratio in Neon Resuspension Buffer R (Thermo Fisher Scientific) and incubated it at room temperature for 10 min to form Cas9 ribonucleoprotein (RNP) complexes. SNU16m1 cells were collected and washed twice with 1× PBS before being resuspended in Buffer R with Cas9 RNPs for a final concentration of 300,000 cells per 10 µl Neon reaction with 0.71 µM Cas9 complexes. Transfection

was performed using the Neon Transfection System (Thermo Fisher Scientific, MPK5000) according to the manufacturer's protocol using 10 µl tips with the following parameters: 1,400 V, 20 ms⁻¹, 2 pulses. Three Neon reactions per guide condition were combined, resulting in 900,000 cells for either the control or *CIP2A*-knockout genotype.

WGS

WGS libraries were prepared by DNA tagmentation. We first transposed genomic DNA with Tn5 transposase produced as previously described⁵⁸, in a 50 µl reaction with TD buffer⁵⁹, 50 ng DNA and 1 µl transposase. The reaction was performed at 50 °C for 5 min, and transposed DNA was purified using the MinElute PCR Purification Kit (Qiagen, 28006). Libraries were generated by 5–7 rounds of PCR amplification using the NEBNext High-Fidelity 2× PCR Master Mix (NEB, M0541L), purified using SPRIselect reagent kit (Beckman Coulter, B23317) with double size selection (0.8× right, 1.2× left) and sequenced on the Illumina NextSeq 550 or the Illumina NovaSeq 6000 platform. Reads were trimmed of adapter content with Trimmomatic⁶⁰ (v.0.39), aligned to the hg19 genome using BWA MEM⁶¹ (0.7.17-r1188) and PCR duplicates were removed using Picard's MarkDuplicates (v.2.25.3). WGS data from bulk SNU16 cells were previously generated (SRR530826, Genome Research Foundation).

Analysis of ecDNA sequences in TCGA patient tumours

We performed ecDNA detection based on bulk WGS data from TCGA using the AmpliconArchitect (AA) method for genomic focal amplification analysis. The outputs of this method were previously published¹⁹. In brief, this approach for detecting ecDNA uses three general steps which are wrapped into a workflow we call AmpliconSuite-pipeline (<https://github.com/AmpliconSuite/AmpliconSuite-pipeline>, v.1.1.1). First, given a BAM file, the analysis pipeline performs detection of seed regions where copy-number amplifications exist (CN > 4.5 and size between 10 kb and 10 Mb). Second, AA performs joint analysis of copy number and breakpoint detection in the focally amplified regions, forming a copy-number aware local genome graph. AA extracts paths representing genome structures and substructures from this graph that explains the changes in copy number. Last, a rule-based classification is performed using AmpliconClassifier (AC)⁶², based on the paths extracted by AA to predict the mode of focal amplification. This includes assessing structural variant types, segment copy numbers and the structure of the genome paths extracted by AA. Moreover, AC identifies ecDNA cycles based on criteria such as cyclic path length and copy number, providing a comprehensive classification system for amplicons on the basis of their structural characteristics. For example, if the changes in copy number are explained predominantly by one or more circular genome paths featuring a structural variant enclosing them with a head-to-tail circularization, this is consistent with an ecDNA mode of amplification, whereas a breakage-fusion-bridge genome structure contains multiple foldbacks and multiple genomic segments arranged in a palindrome. The complete classification criteria and description of the AC tool are available in the supplementary information of ref. 62.

We used AA (v.1.0) outputs from a previous study¹⁹, and classified focal amplifications types present in these outputs using AC (v.0.4.14) with the '--filter_similar' flag set and otherwise the default settings. The '--filter_similar' option removes probable false-positive focal amplification calls that contain far greater-than-expected levels of overlapping structural variants and shared genomic boundaries between ecDNAs of unrelated samples. In brief, AC scores the structural similarity of focal amplifications. These scores consider both genomic interval overlap and shared breakpoint junctions, with breakpoints deemed to be shared if their total distance is less than a specified threshold (default = 250 bp). Moreover, AC computes similarity scores for amplicons from unrelated origins, establishing a background null distribution for comparison. The tool uses a β -distribution model to fit the empirical null distribution, providing estimation of statistical

significance of the similarity score. Out of 8,810 AA amplicons in the ref. 19 TCGA dataset, 45 candidate focal amplifications were removed by this filter.

To predict the distinct number of ecDNA species present in a sample, we used the genome intervals reported by AC for each focal amplification. AC determines the number of distinct, genomically non-overlapping ecDNA species present by clustering ecDNA genome intervals if those regions are connected by structural variants or the boundaries of the regions are within 500 kb. If intervals do not meet this criteria, AC predicts them as being unconnected and reports them as separate ecDNA species. AC uses a list of oncogenes that combines genes in the OGene database (<https://pubmed.ncbi.nlm.nih.gov/28162959/>) and COSMIC (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6450507/>).

Paired scATAC-seq and scRNA-seq library generation

Single-cell paired RNA-seq and ATAC-seq libraries were generated on the 10x Chromium Single-Cell Multiome ATAC + Gene Expression platform according to the manufacturer's protocol and sequenced on an Illumina NovaSeq 6000 system. Data for COLO 320DM were generated previously⁴ and published under Gene Expression Omnibus (GEO) accession GSE159986.

Paired scATAC-seq and scRNA-seq analysis

A custom reference package for hg19 was created using cellranger-arc mkref (10x Genomics, v.1.0.0). The single-cell paired RNA-seq and ATAC-seq reads were aligned to the hg19 reference genome using cellranger-arc count (10x Genomics, v.1.0.0).

Subsequent analyses on RNA were performed using Seurat (v.3.2.3)⁶³, and those on ATAC-seq were performed using ArchR (v.1.0.1)⁶⁴. Cells with more than 200 unique RNA features, less than 20% mitochondrial RNA reads and less than 50,000 total RNA reads were retained for further analyses. Doublets were removed using ArchR. Raw RNA counts were log-normalized using Seurat's NormalizeData function and scaled using the ScaleData function. Dimensionality reduction for the ATAC-seq data was performed using Iterative Latent Semantic Indexing (LSI) with the addIterativeLSI function in ArchR.

We next calculated amplicon copy numbers based on background ATAC-seq signals as we previously described and validated^{4,32}. In brief, we determined read counts in large intervals across the genome using a sliding window of 3 Mb moving in 1 Mb increments across the reference genome. Genomic regions with known mapping artifacts were filtered out using the ENCODE hg19 blacklist. For each interval, insertions per bp were calculated and compared to 100 of its nearest neighbours with matched GC nucleotide content. The mean $\log_2[\text{fold change}]$ was computed for each interval. On the basis of a diploid genome, copy numbers were calculated using the formula $CN = 2 \times 2^{\log_2[FC]}$, where CN denotes copy number and FC denotes mean fold change compared with neighbouring intervals. To query the copy numbers of a gene, we obtained all genomic intervals that overlapped with the annotated gene sequence and computed the mean copy number of those intervals.

For analyses presented in Extended Data Fig. 5a–c, we inferred cell cycle stage from each cell's RNA-seq data using the CellCycleScoring function in Seurat and the gene sets for S and G2M phases included in the Seurat package. Copy-number correlations were then evaluated for cells grouped by their inferred cell cycle phase: G1, S, or G2M.

scCircle-seq analysis

TR14 scCircle-seq data were previously generated⁶⁵ and deposited at the European Genome-Phenome Archive (EGA) under accession number EGAS00001007026. A detailed description of the single-cell extrachromosomal circular DNA and transcriptome sequencing (scEC&T-seq) protocol is available at Nature Protocol Exchange (<https://doi.org/10.21203/rs.3.pex-2180/v1>)⁶⁶. Single cells were sorted, separation of genomic DNA and mRNA was performed by G&T-seq⁶⁷ and genomic DNA

of single cells was subjected to exonuclease digestion and rolling-circle amplification as described previously⁶⁵.

The processing of scCircle-seq reads is described in detail previously⁶⁵. In brief, scCircle-seq sequencing reads were 3' trimmed for quality using Trim Galore (v.0.6.4)⁶⁸, and adapter sequences with reads shorter than 20 nucleotides were removed. The alignment of reads to the human reference assembly hg19 was performed using BWA MEM (v.0.7.15) with the default parameters⁶⁹. PCR and optical duplicates were removed using Picard (v.2.16.0). Sequencing coverage across mitochondrial DNA was used as an internal control to evaluate circular DNA enrichment. Cells that exhibited less than 10 reads per bp sequence-read depth over mitochondrial DNA or less than 85% genomic bases captured in mitochondrial DNA were excluded from further analyses⁶⁵.

Read counts from scCircle-seq BAM files were quantified in 1 kb bins across TR14 ecDNA regions (*MYNC*, *CDK4*, *MDM2*) as defined by ecDNA reconstruction analyses in TR14 bulk populations described previously⁴. To account for differences in sequencing depth among cells, read counts were normalized to library size.

Analysis of copy-number correlations of amplified oncogenes in human tumour samples

Copy numbers computed for single cells using scATAC-seq as described above (see the 'Paired scATAC-seq and scRNA-seq analysis' section) were used to devise a statistical approach for predicting ecDNA. We reasoned that, due to the random segregation of individual ecDNA molecules, ecDNA focal amplifications would be characterized by not only elevated mean copy number but also inflated copy-number variance. Indeed, classifying amplifications with a mean copy number of ≥ 4 and variance/mean ratio of ≥ 2.5 specifically classified only known ecDNAs in validated cell lines (Extended Data Fig. 4a).

We applied this statistical approach to a curated dataset of 41 tumours (from triple-negative breast cancer (TNBC), high-grade serous ovarian cancer (HGSC) and glioblastoma) with publicly available scATAC-seq or scDNA-seq data^{34–36}. For TNBC and HGSC tumours profiled with scDNA-seq data in ref. 35, we used the author-provided single-cell copy numbers available on Zenodo (<https://doi.org/10.5281/zenodo.6998936>). Processed scATAC-seq data for glioblastoma samples were obtained from ref. 34 and ref. 36 (GEO accession number GSE163655), and copy numbers were computed as described above (see the 'Paired scATAC-seq and scRNA-seq analysis' section) in 3 Mb genomic windows. Putative ecDNAs were predicted using the decision rule determined from validated cell lines, and copy numbers were determined for oncogenes by averaging copy numbers of windows overlapping with the oncogene of interest. Copy-number correlations were computed across oncogenes, only considering cells where the oncogene was amplified with a copy-number ≥ 4 .

ecDNA isolation by CRISPR–CATCH

Molecular isolation of ecDNA by CRISPR–CATCH was performed as previously described¹⁸. In brief, molten 1% certified low-melting-point agarose (Bio-Rad, 1613112) in PBS was equilibrated to 45 °C. In total, 1 million cells were pelleted per condition, washed twice with cold 1× PBS, resuspended in 30 μ l PBS and briefly heated to 37 °C. Then, 30 μ l agarose solution was added to cells, mixed, transferred to a plug mould (Bio-Rad, 1703713) and incubated on ice for 10 min. Solid agarose plugs containing cells were ejected into 1.5 ml Eppendorf tubes, suspended in buffer SDE (1% SDS, 25 mM EDTA at pH 8.0) and placed onto a shaker for 10 min. The buffer was removed and buffer ES (1% *N*-lauroylsarcosine sodium salt solution, 25 mM EDTA at pH 8.0, 50 μ g ml^{−1} proteinase K) was added. Agarose plugs were incubated in buffer ES at 50 °C overnight. The next day, proteinase K was inactivated with 25 mM EDTA with 1 mM PMSF for 1 h at room temperature with shaking. Plugs were then treated with RNase A (1 mg ml^{−1}) in 25 mM EDTA for 30 min at 37 °C and washed with 25 mM EDTA with a 5 min